

Quantum Gravitation

Lecture 8

Midterm Review

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$checksum = hash(C_{66}, grid, basis)$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$\Pi^{(2)} : \Pi^{(2)} = 5$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$\Delta E_{num} < 0.01 \sigma_{registry}$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Midterm formula card; Q-Medium exam snapshot guide; Weeks 1–7 run records